WEEK 01 — Introduction to Computer Organization and Computer Systems

Complete Notes
Conceptual + Theory
Numericals

PART 1: Introduction to Computer Systems

1. What is a Computer?

Definition (Exam):

A computer is an electronic programmable device that accepts data as input, processes it according to given instructions, stores data and produces meaningful information as output.

Explanation

Computer basically ek electronic machine hai jo user se raw data leti hai, us par operations perform karti hai aur meaningful result provide karti hai.

For example, agar hum computer ko 20 + 30 calculate karne ko kahein:

- Input: 20 and 30
- Processing: Addition
- Output: 50
- Storage: Result ko memory mein save bhi kiya ja sakta hai.

Basic Functions of Computer

Information Processing Cycle
in.pinterest.com

1. Input

Input is the process of entering data and instructions into a computer.

Examples: Keyboard, mouse, scanner.

2. Processing

Processing is the operation performed on input data to convert it into meaningful information.

Example: Adding two numbers.

3. Output

Output is the result produced by a computer after processing.

Examples: Monitor, printer, speakers.

4. Storage

Storage is the process of saving data, instructions and results for future use.

Examples: RAM, SSD, HDD.

Conceptual Question:

Q: What is the difference between data and information?

- Data: Raw facts and figures that have not been processed.
- Information: Processed data that has meaning and is useful.

Example:

- Data: 70, 80, 90
- Information: Average marks = 80.

2. What is Computer Organization?

Definition (Exam):

Computer Organization refers to the internal arrangement and operational structure of hardware components that work together to execute computer instructions.

Explanation

Computer organization explain karti hai ke computer ke different hardware components internally kis tarah connected hain aur mil kar kaam kaise karte hain.

For example:

- CPU data ko process kaise karta hai?
- ALU addition kaise perform karta hai?
- Memory CPU ke saath communicate kaise karti hai?
- Control Unit different components ko control kaise karti hai?
- Data buses information transfer kaise karti hain?

Ye sab Computer Organization ka part hain.

Main Components of Computer Organization

1. CPU
2. ALU
3. Control Unit
4. Registers
5. Memory
6. Input/Output Units
7. System Bus

Example:

Jab CPU ko 5 + 10 calculate karna hota hai, to Control Unit instruction ko control karti hai, registers data hold karte hain aur ALU addition perform karti hai.

3. What is Computer Architecture?

Definition (Exam):

Computer Architecture refers to the functional design and characteristics of a computer system that are visible to the programmer.

Explanation

Computer Architecture batati hai ke computer programmer ko kya facilities provide karta hai aur computer instructions kis tarah defined hoti hain.

It includes:

1. Instruction Set Architecture (ISA)
2. Data types
3. Addressing modes
4. Registers visible to programmers
5. Memory addressing
6. Instruction formats

Example:

Suppose a processor supports instructions such as:

- ADD
- SUB
- MOV
- MUL

These instructions are part of its Instruction Set Architecture.

4. Difference Between Computer Architecture and Computer Organization

Very Important — Long Question

Computer Architecture	Computer Organization
Describes what a computer does.	Describes how a computer works.
Deals with functional behavior.	Deals with hardware implementation.
Programmer-oriented.	Hardware-oriented.
Defines instruction sets.	Explains instruction execution hardware.
Includes addressing modes.	Includes buses and control signals.
Defines data types.	Explains how data is handled internally.

Practical Example

Suppose two computers support the same instruction set and produce the same results.

However:

- Computer A uses a different internal hardware design.
- Computer B uses another internal hardware design.

Both may have the same computer architecture but different computer organizations.

Why?

Because both computers perform the same functions, but their internal hardware implementation can be different.

Important Conceptual Questions

Q1. Can two computers have the same architecture but different organization?

Yes. Two computers can support the same instruction set and programmer-visible features but use different internal hardware designs and implementations.

Q2. Which is more concerned with hardware implementation?

Computer Organization.

Q3. Which is more visible to programmers?

Computer Architecture.

Q4. Does architecture describe how circuits are physically connected?

Generally, no. Internal hardware connections and implementation belong to Computer Organization.

5. Importance of Computer Organization for IT Students

Definition:

Computer Organization helps students understand how different hardware components interact to execute instructions and process data.

Importance

1. Understanding CPU: It helps students understand how the processor executes instructions.
2. Memory Management: It explains how data is stored and retrieved from memory.
3. Programming: It provides knowledge about low-level programming and machine instructions.
4. Performance: It helps understand how hardware affects computer performance.
5. Operating Systems: It provides a foundation for understanding processes, memory and hardware interaction.
6. Assembly Language: It helps students understand registers, instructions and processor operations.
7. Troubleshooting: It helps identify hardware-related performance and operational issues.

Conceptual Question:

Q: Why should a software engineer study Computer Organization?

Answer: A software engineer should study Computer Organization to understand how software instructions are executed by hardware, how memory is accessed, how the CPU processes data and how hardware architecture affects software performance.

PART 2: Basic Components of Computer System

6. Block Diagram of Computer System

Ye diagram Week 1 ka sab se important conceptual diagram hai.

PPT - A Simple Computer consists of a Processor (CPU-Central Processing Unit), Memory, and I/O PowerPoint Presentation - ID:330894
slideserve.com

A computer system consists of the following major functional units:

1. Input Unit
2. Central Processing Unit (CPU)
3. Memory Unit
4. Output Unit

CPU itself consists of:

- ALU
- Control Unit
- Registers

Let's understand every component.

7. Input Unit

Definition:

The Input Unit is responsible for accepting data and instructions from the user and converting them into a form that can be understood and processed by the computer.

Functions

1. Accepts data from the user.
2. Accepts instructions.
3. Converts input into machine-readable form.
4. Sends data and instructions to the computer for processing.

Examples

- Keyboard
- Mouse
- Scanner
- Microphone
- Touchscreen

Conceptual Question:

Q: Why is an input unit necessary?

Because a computer needs data and instructions to perform operations. The input unit provides this information to the computer.

8. Output Unit

Definition:

The Output Unit is responsible for presenting processed information to the user in a human-readable form.

Functions

1. Receives processed results.
2. Converts results into understandable form.
3. Displays or produces information.

Examples

- Monitor
- Printer
- Speaker
- Projector

Conceptual Question:

Q: What is the difference between input and output devices?

Input devices send data to the computer, whereas output devices receive processed information from the computer and present it to the user.

9. Memory Unit

Memory Hierarchy in Computer Architecture, Levels and Balancing Speed
testbook.com

Definition:

Memory is the storage component of a computer system that stores data, instructions and processing results.

Functions of Memory

1. Stores input data.
2. Stores program instructions.
3. Stores intermediate results.

4. Stores final results.
5. Supplies data and instructions to the CPU when required.

Types of Memory

Memory is mainly divided into two categories:

A. Primary Memory

Primary memory is directly accessible by the CPU and stores data and instructions currently needed by the system.

Examples:

- RAM
- ROM

RAM (Random Access Memory)

Definition: RAM is a volatile primary memory that temporarily stores data and instructions currently being used by the CPU.

Important points:

- It is volatile.
- Data is lost when power is turned off.
- It supports read and write operations.
- It is faster than secondary storage.

Example: When you open Chrome, its running data is loaded into RAM.

ROM (Read Only Memory)

Definition: ROM is a non-volatile memory that stores permanent instructions required to start and operate a computer.

Important points:

- It is non-volatile.
- Data is retained without power.
- It commonly stores firmware.
- It is not normally used as general-purpose working memory.

Example: Firmware used during computer startup.

B. Secondary Memory

Definition: Secondary memory is non-volatile storage used to store data and programs permanently or for long-term use.

Examples:

- Hard Disk Drive (HDD)
- Solid State Drive (SSD)
- USB Flash Drive
- Memory Card

RAM vs ROM vs Secondary Memory

RAM	ROM	Secondary Memory
Volatile	Non-volatile	Non-volatile
Temporary storage	Stores firmware	Long-term storage
Read and write	Mostly read-oriented	Read and write
Fast working memory	Stores startup instructions	Stores files and programs
Example: DDR RAM	Example: Firmware ROM	Example: SSD

Conceptual Questions

Q1. Why is RAM called volatile memory?

Because RAM loses its stored information when the power supply is turned off.

Q2. Why is ROM called non-volatile memory?

Because it retains its information even when the power is turned off.

Q3. Is SSD primary or secondary memory?

SSD is secondary storage.

Q4. Is RAM faster than SSD?

Yes. RAM is generally much faster than SSD for active data access.

Q5. Why does CPU need RAM?

CPU uses RAM to access currently required instructions and data quickly instead of repeatedly retrieving them from slower secondary storage.

PART 3: CPU and Its Components

10. What is CPU?

Paper 1 Lesson 2.3 - CPU Architecture | Graded IB Computer Science Notes
graded-cs-resources.github.io

Definition:

The Central Processing Unit (CPU) is the main processing component of a computer that executes instructions, performs arithmetic and logical operations, and controls other components.

CPU ko computer ka brain kaha jata hai because it controls and processes most of the computer's operations.

Main Functions of CPU

1. Fetches instructions from memory.
2. Decodes instructions.
3. Executes instructions.
4. Performs arithmetic operations.
5. Performs logical operations.
6. Controls data movement between components.

Main Components of CPU

1. Arithmetic Logic Unit (ALU)
2. Control Unit (CU)
3. Registers

11. Arithmetic Logic Unit (ALU)

Definition:

The Arithmetic Logic Unit (ALU) is a part of the CPU that performs arithmetic, logical and comparison operations on data.

ALU ka main kaam calculations aur logical operations perform karna hai.

A. Arithmetic Operations

Arithmetic operations are mathematical calculations performed on numerical data.

Examples:

Operation	Example

Addition	10 + 5 = 15
Subtraction	10 - 5 = 5
Multiplication	10 × 5 = 50
Division	10 ÷ 5 = 2
Increment	5 + 1 = 6
Decrement	5 - 1 = 4

B. Logical Operations

Logical operations work on binary values (0 and 1).

Important operations:

AND Operation

Output is 1 only when both inputs are 1.

A	B	A AND B
0	0	0
0	1	0
1	0	0
1	1	1

OR Operation

Output is 1 when at least one input is 1.

A	B	A OR B
0	0	0
0	1	1
1	0	1
1	1	1

NOT Operation

NOT reverses the input.

A	NOT A
0	1
1	0

XOR Operation

Output is 1 when both inputs are different.

A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

C. Comparison Operations

ALU also performs comparison operations.

Examples:

- 10 > 5
- 5 < 10
- 5 = 5

The result of comparison may affect status flags in the CPU.

Important Conceptual Questions

Q1. Does ALU control the entire computer?

No. ALU performs arithmetic and logical operations. The Control Unit coordinates the overall operations.

Q2. Does ALU store data permanently?

No. ALU performs operations on data. Registers temporarily hold the operands and results.

Q3. Which component performs subtraction?

ALU.

12. Control Unit (CU)

Definition:

The Control Unit is a component of the CPU that directs and coordinates the operations of all computer components by generating control signals.

Explanation

Control Unit ko computer ka manager samajh lo.

Ye khud calculations perform nahi karti. Instead, ye decide karti hai:

- Which instruction should be executed?
- Which component should work?
- When should data be transferred?
- When should ALU perform an operation?
- When should memory be read or written?

Functions of Control Unit

1. Fetches instructions from memory.
2. Decodes instructions.
3. Generates control signals.
4. Controls the execution of instructions.
5. Coordinates ALU, registers and memory.
6. Manages the sequence of operations.

Example

Suppose CPU needs to execute:

\[5+3 \]

What happens?

1. Control Unit fetches the ADD instruction.
2. It decodes the instruction.
3. It identifies the required operands.
4. It sends control signals to the registers and ALU.
5. ALU performs addition.
6. The result is stored in a register.

ALU vs Control Unit

ALU	Control Unit
Performs arithmetic operations.	Controls computer operations.
Performs logical operations.	Generates control signals.
Processes data.	Coordinates data movement.
Calculates results.	Directs instruction execution.

Remember:

ALU = Worker (performs calculations).

CU = Manager (controls the work).

13. Registers

General Purpose Registers.pptx
slideshare.net

Definition:

Registers are small, high-speed storage locations inside the CPU that temporarily hold data, instructions, addresses and intermediate results during processing.

Explanation

Registers CPU ke andar small storage locations hoti hain.

Jab CPU ko kisi data par operation perform karna hota hai, to required data registers mein temporarily rakha ja sakta hai.

Why are registers needed?

1. To store data temporarily.
2. To hold instructions.
3. To store memory addresses.
4. To hold intermediate results.
5. To provide fast access to required information.

Types of Registers

At a basic level, registers are divided into two categories.

A. General-Purpose Registers

These registers are used to store temporary data and intermediate results during program execution.

Examples in 8086:

- AX
- BX
- CX
- DX

B. Special-Purpose Registers

These registers perform specific functions in the processor.

Examples:

- Program Counter (PC)
- Instruction Register (IR)
- Stack Pointer (SP)
- Memory Address Register (MAR)
- Memory Data Register (MDR)

Note: 8086 ke specific registers aur unke functions hum Week 7 mein detail se cover karenge.

Registers vs RAM

Registers	RAM
Located inside CPU.	Main memory outside CPU core.
Very small capacity.	Much larger capacity.
Extremely fast.	Slower than registers.
Hold currently required data.	Holds active programs and data.
Directly used during CPU operations.	Supplies data to CPU.

Conceptual Question:

Q: Why are registers faster than RAM?

Registers are located inside the CPU and are designed for very fast access during instruction execution, whereas RAM is the main memory and requires memory access operations.

PART 4: System Bus

14. What is a Bus?

Definition:

A Bus is a communication pathway used to transfer data, addresses and control signals between different components of a computer system.

Explanation

Computer ke CPU, memory aur input/output devices ko aapas mein communicate karna hota hai.

For example:

- CPU ko memory se data chahiye.
- CPU ko memory mein data save karna hai.
- CPU ko I/O device se information receive karni hai.

In sab ke liye buses use hoti hain.

15. System Bus

Definition:

A System Bus is a collection of communication pathways that connects the CPU, memory and input/output devices to transfer data, addresses and control signals.

System Bus ke three main types hain:

1. Data Bus
2. Address Bus
3. Control Bus

Chapter 2: A Top-Level View of Computer Function & Interconnection | by Nik Aishah | BICS1302-Study Reflection | Medium
medium.com

16. Data Bus

Definition:

A Data Bus is a communication pathway that transfers actual data between the CPU, memory and input/output devices.

Characteristics

1. It carries actual data.
2. It is generally bidirectional.
3. Its width determines how many bits can be transferred at a time.

What is Bidirectional?

Bidirectional means data can travel in both directions.

Example:

- CPU → Memory
- Memory → CPU

Example

If a computer has a 32-bit Data Bus, it can transfer:

$$ 32\ bits = \frac{32}{8}=4\ Bytes $$

at a time.

17. Address Bus

Definition:

An Address Bus is a communication pathway used by the CPU to send the address of a memory location or I/O location that needs to be accessed.

Characteristics

1. It carries addresses, not actual data.
2. It is generally unidirectional from CPU.
3. Its width determines the number of addressable locations.

What is Unidirectional?

Unidirectional means information travels in only one direction.

Example:

CPU → Memory (address).

Important Formula

$$ \boxed{\text{Addressable Locations}=2^n} $$

Where n = Number of address bits.

Example:

If Address Bus = 8 bits:

$$ 2^8=256 $$

So it can address 256 different locations.

18. Control Bus

Definition:

A Control Bus is a communication pathway that carries control and timing signals between the CPU, memory and input/output devices.

Characteristics

1. It carries control signals.
2. It coordinates computer operations.
3. Signals can travel in different directions depending on their purpose.

Important Control Signals

Signal	Function
READ	Requests data from memory.
WRITE	Requests memory to store data.
INTERRUPT	Requests CPU attention.
CLOCK	Synchronizes operations.
RESET	Initializes or restarts processor operation.

Example

When CPU wants to read data from memory:

- Address Bus sends the address.
- Control Bus sends READ signal.
- Data Bus transfers the requested data.

Ye teeno buses mil kar kaam karti hain.

19. Difference Between Data Bus, Address Bus and Control Bus

Important Long Question

Feature	Data Bus	Address Bus	Control Bus
Purpose	Transfers data	Transfers addresses	Transfers control signals
Direction	Bidirectional	Generally unidirectional	Depends on signal
Carries	Actual information	Memory location	Read, Write, Clock etc.
Width	Determines data per transfer	Determines addressable locations	Depends on control design
Example	32-bit	20-bit	Read/Write signals

Most Important Conceptual Question

Q: Why do we need three different buses? Why not use only one?

Answer:

The three buses perform different functions. The Data Bus transfers actual information, the Address Bus identifies the location where information should be read or written, and the Control Bus coordinates the operation through signals such as READ and WRITE. Separating these functions makes communication between computer components organized and efficient.

PART 5: CPU-Memory Communication

20. What is CPU-Memory Communication?

Definition:

CPU-Memory Communication is the process of exchanging data, instructions and addresses between the CPU and main memory through the system bus.

CPU aur memory ko continuously communicate karna hota hai because CPU instructions aur data memory se retrieve karta hai aur results wapas memory mein store bhi kar sakta hai.

There are two main operations:

1. Memory Read
2. Memory Write

21. Memory Read Operation

Definition:

A Memory Read Operation is the process in which the CPU retrieves data or instructions from a specified memory location.

Fetch-execute Cycle | PPT
slideshare.net

Step-by-step working

Suppose CPU wants to read data from memory location 200.

Step 1: CPU places address 200 on the Address Bus.

Step 2: CPU sends a READ signal through the Control Bus.

Step 3: Memory identifies location 200.

Step 4: Memory retrieves the data stored at location 200.

Step 5: Memory sends the requested data through the Data Bus.

Step 6: CPU receives the data and stores it in an appropriate register.

Example:

Memory location 200 contains 50.

CPU requests location 200.

Result received by CPU = 50.

22. Memory Write Operation

Definition:

A Memory Write Operation is the process in which the CPU stores data into a specified memory location.

PPT - ECP2036 Microprocessor and Interfacing PowerPoint Presentation, free download - ID:9710280
slideserve.com

Step-by-step working

Suppose CPU wants to store 100 at memory location 300.

Step 1: CPU places address 300 on the Address Bus.

Step 2: CPU places data 100 on the Data Bus.

Step 3: CPU sends a WRITE signal through the Control Bus.

Step 4: Memory identifies location 300.

Step 5: Memory stores 100 at location 300.

Result:

Memory[300] = 100.

Difference Between Read and Write

Memory Read	Memory Write
Retrieves data from memory.	Stores data in memory.
Memory sends data to CPU.	CPU sends data to memory.
READ signal is used.	WRITE signal is used.
Data travels Memory → CPU.	Data travels CPU → Memory.

Conceptual Questions

Q1. Which bus carries the memory address?

Address Bus.

Q2. Which bus carries actual data?

Data Bus.

Q3. Which bus carries READ and WRITE signals?

Control Bus.

Q4. During a read operation, who provides the data?

Memory provides the data to CPU.

Q5. During a write operation, who provides the data?

CPU provides the data to memory.

PART 6: Numerical Preparation — Week 1

Ab important part. Tumhari mam ne numerical questions ka kaha hai, so ye concepts aur formulas achi tarah samajh lo.

23. Basic Units of Computer Memory

Ye units yaad hona bohat zaroori hain.

Unit	Equivalent
1 Bit	0 or 1
1 Nibble	4 Bits

1 Byte	8 Bits
1 KB	1024 Bytes
1 MB	1024 KB
1 GB	1024 MB
1 TB	1024 GB

Remember:

\[1\ Byte=8\ Bits \]
\[1\ KB=2^{10}\ Bytes \]
\[1\ MB=2^{20}\ Bytes \]
\[1\ GB=2^{30}\ Bytes \]

24. Address Bus Numerical

Formula:

\[\boxed{2^n=\text{Number of addressable locations}} \]

Where n = Address Bus width.

Question 1

A computer has an 8-bit address bus. Calculate the number of memory locations it can address.

Solution:

Given:

\[n=8 \]

Formula:

\[2^n \]

Putting values:

\[2^8=256 \]
Answer: 256 locations

Question 2

A computer has a 16-bit address bus. Find its addressable memory capacity if each location stores 1 byte.

Solution:

$$2^{16}=65,536\ \text{Locations}$$

Each location = 1 Byte.

$$65,536\ \text{Bytes}$$

Convert into KB:

$$\frac{65,536}{1024}=64\ \text{KB}$$
Answer: 64 KB

Question 3

A computer has a 20-bit address bus. Calculate its maximum addressable memory.

$$2^{20}=1,048,576\ \text{Bytes}$$
$$\frac{1,048,576}{1024}=1024\ \text{KB}$$
$$=1\ \text{MB}$$
Answer: 1 MB

Question 4

A computer has 4 MB of byte-addressable memory. Find the minimum number of address bits required.

Solution:

Given:

$$4\ \text{MB}=4\times1024\times1024$$
$$=4,194,304\ \text{Bytes}$$

Now:

$$2^{22}=4,194,304$$

Therefore:

$$n=22$$
Answer: 22 address bits

25. Data Bus Numerical

Formula:

$$\boxed{\text{Bytes per transfer}=\frac{\text{Data Bus width}}{8}}$$

Question 1

A computer has a 32-bit data bus. How many bytes can it transfer at a time?

$$32/8=4$$
Answer: 4 Bytes

Question 2

A computer has a 64-bit data bus. Calculate the data transfer capacity in bytes.

$$64/8=8$$
Answer: 8 Bytes

Question 3

How many bits are transferred in 5 transfers using a 16-bit data bus?

Each transfer = 16 bits.

Total:

$$16\times5=80\ \text{Bits}$$
Answer: 80 Bits

26. Memory Capacity Numerical

Question:

A computer has 1024 memory locations, and each location stores 16 bits. Calculate the total memory capacity in bytes.

Step 1: Number of locations = 1024.

Step 2: Each location = 16 bits.

Total bits:

$$1024\times16=16,384\ \text{Bits}$$

Step 3: Convert bits into bytes.

$$ 16,384/8=2048\ \text{Bytes} $$

Step 4: Convert into KB.

$$ 2048/1024=2\ \text{KB} $$
Answer: 2 KB

27. Find Address Bus Width from Memory Capacity

Question:

A computer has 64 KB of byte-addressable memory. Find the required address bus width.

Solution:

$$ 64\ \text{KB}=64\times1024 $$
$$ =65,536\ \text{Bytes} $$

Now:

$$ 2^{16}=65,536 $$

Therefore:

$$ n=16 $$
Answer: 16-bit Address Bus

PART 7: Most Important Conceptual Questions

Ye questions prepare karna because mam conceptual form mein bhi pooch sakti hain.

Question	Key concept
Why is CPU called the brain of the computer?	Executes and controls operations
Why are registers faster than RAM?	Located inside CPU
Why do we need RAM?	Holds active data and instructions
Why is ROM non-volatile?	Retains data without power
Can CPU work without memory?	Needs storage for instructions and data
What happens if the Address Bus width increases?	Addressable locations increase
What happens if Data Bus width increases?	More bits can transfer per operation

Why is the Control Bus necessary?	Coordinates operations
Can ALU work without CU coordination?	Needs control signals to perform the required operation
Why are buses important?	Enable communication between components
What happens during Memory Read?	Data moves from memory to CPU
What happens during Memory Write?	Data moves from CPU to memory

PART 8: Important Long Questions

Inko written practice bhi karna.

1. Explain the basic functional units of a computer system with a neat diagram.
2. Define Computer Architecture and Computer Organization. Differentiate between them.
3. Explain CPU and its major components in detail.
4. Explain the functions of ALU and Control Unit.
5. What are registers? Explain their importance and types.
6. Explain the system bus and its three major types.
7. Differentiate between Data Bus, Address Bus and Control Bus.
8. Explain CPU-Memory Read and Write operations with diagrams.
9. Explain the memory hierarchy and differentiate between primary and secondary memory.
10. Explain the importance of Computer Organization for IT students.

PART 9: Practice Numericals — Solve Yourself

Ab ye 6 questions khud solve karna. Ye tumhare Week 1 ke important numerical patterns cover karte hain.

Week 1 Numerical & Conceptual Test

0/6 attempted

Q1. A computer has a 12-bit address bus. How many memory locations can it address?

1024
2048
4096
8192

Q2. A 64-bit data bus can transfer how many bytes at a time?

4 Bytes

8 Bytes
16 Bytes
64 Bytes

Q3. A computer has 256 memory locations, each storing 8 bits. What is its total capacity?

128 Bytes
256 Bytes
512 Bytes
1024 Bytes

Q4. What is the address bus width required for 1 MB byte-addressable memory?

16 bits
18 bits
20 bits
24 bits

Q5. A 32-bit data bus performs 10 transfers. How many total bytes are transferred?

20 Bytes
32 Bytes
40 Bytes
64 Bytes

Q6. Which bus carries the WRITE signal?

Data Bus
Address Bus
Control Bus
None
Check My Answers

Final Revision — Week 1 Formula Sheet

Is table ko paper se pehle zaroor revise karna.

Topic	Formula
Addressable locations	\(2^n\)
Memory capacity	Locations × Bits per location
Bits to Bytes	Bits ÷ 8
Bytes to Bits	Bytes × 8
Bytes to KB	Bytes ÷ 1024
KB to Bytes	KB × 1024

Bytes per transfer	Data Bus width ÷ 8
Required address bits	$\log_2(\text{Locations})$

Important assumptions:

- 2^n gives addressable locations, not automatically bytes.
- Agar each location 1 byte store karti hai, then capacity in bytes = 2^n.
- Agar each location 16 bits store karti hai, then total capacity = $2^n \times 16$ bits.
- Memory sizes in these examples use binary units (1 KB = 1024 bytes).

Meri recommendation for tomorrow's paper

Week 1 mein in 5 areas ko bilkul strong kar lo:

1. Computer Architecture vs Computer Organization.
2. CPU, ALU, CU and Registers.
3. Data Bus, Address Bus and Control Bus.
4. CPU-Memory Read and Write operations.
5. Address Bus width, Data Bus width and Memory Capacity numericals.